

Optimal Block Time for AMM Liquidity Providers under Jump-Diffusion Prices

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Nils Bundi

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The case for faster blocks

Two live conversations about how fast a chain should be

Arbitrum Research — “The power of faster blocks” (Felten, May 2024)

“Arbitrum’s 250 millisecond block time leads to 65% lower arbitrage loss compared to a 2 second block time.”

research.arbitrum.io/t/the-power-of-faster-blocks/9609 ↗

EIP-7782 “Reduce Block Latency” — 12 s → 6 s, drafted for Glamsterdam

“More frequent blocks decrease LVR (Loss Versus Rebalancing), which improves the economics for liquidity providers. More liquidity means lower spreads.”

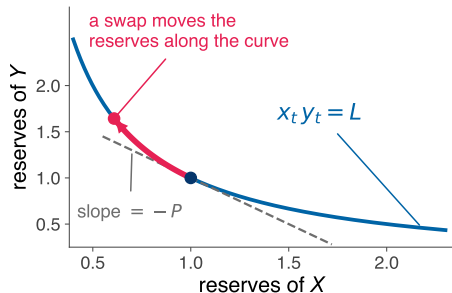
eips.ethereum.org/EIPS/eip-7782 ↗

Both rest on the same result. Both, on the numbers below, overstate the gain — by roughly a factor of two.

What an automated market maker is

A smart contract, or pool, holds reserves x of X and y of Y . Anyone may swap against it. There is no order book and nobody quoting.

- The rule is an invariant, $x_t y_t = L$: a swap is any move along that curve, and the curve alone decides what you get.
- The pool's price is the curve's slope, $P = y/x$. Nothing else sets it — the pool cannot see the outside market.
- LPs supply the reserves and earn a swap fee; they never quote or cancel.



Where the loss comes from

An external move leaves the pool's marginal price stale. Arbitrage restores its price, but the trade executes along the convex curve below the new price. The LP bears the difference.

What we know about LVR

Milionis et al. (2022, 2023) named that loss loss-versus-rebalancing (LVR), and shaped how its relationship with block time is understood today.

- Under geometric Brownian motion, a constant-product pool loses at the rate $\sigma^2 V/8$, independent of the block time.
- Add a swap fee γ and Poisson blocks of mean Δt (Milionis et al., 2023): the rate is multiplied by $F(\kappa)$, $\kappa = \gamma/(\sigma\sqrt{\Delta t})$ — the stationary probability that an arriving block offers a *profitable* trade.
- $F(\kappa) \sim \sigma\sqrt{\Delta t}/(\sqrt{2}\gamma) \rightarrow 0$ as $\Delta t \rightarrow 0$. That $\sqrt{\Delta t}$ is where Arbitrum's 65% comes from, and the $\sim 29\%$ implied by halving Ethereum's slot.

The primitive all of this assumes

Milionis et al. assume a *continuous* reference price, and crypto prices are not (Scaillet et al., 2020, Saef et al., 2024). A jump of size J opens a gap of size J whatever the block time.

Contributions

1. **Two LVR channels.** Once the price can jump, the loss separates into a channel the block schedule governs and one it cannot reach (Theorems 1–2).
2. **A jump induced floor.** Faster blocks clear a jump gap sooner, never smaller. So the loss rate stops falling at a strictly positive level (Theorem 3).
3. **A planner's objective.** Blocks cost something to produce. A planner's welfare problem puts the cost in relation with the LVR loss (Theorem 4).
4. **Empirical calibration.** Magnitudes of loss, cost and optimum are computed from six years of ETH/USDT tick data, and Ethereum's issuance.

Headline

At Ethereum's 12 s, LPs lose 471 bp/yr, 125 of it beyond any schedule's reach; the optimum sits near 8 s. So the two proposals I opened with are worth 35% and 14%, not 65% and 29%.

Two channels, and a floor

Setup

Milionis et al. with one primitive changed: the reference price jumps.

$$\frac{dS_t}{S_{t-}} = \mu dt + \sigma dW_t + (e^{J_t} - 1) dN_t, \quad N \text{ Poisson}(\lambda), \quad J \sim \nu.$$

- CPMM, $x_t y_t = L$, price $P = y/x$, pool value $V(P) = 2\sqrt{LP}$.
- Fee γ in log-price units $\Rightarrow$ no-arbitrage band $\{|\log S - \log P| \leq \gamma\}$.
- Blocks arrive Poisson at rate $1/\Delta t$, so Δt is the *mean* time between them.
- $\text{LVR}(T) = \int_0^T V'(P_{s-}) dP_s - [V(P_T) - V(P_0)]$; rate $\ell = \mathbb{E}_t[d\text{LVR}_t]/dt$.
- ν finite activity with $\int (e^{j/2} - 1)^2 \nu(dj) < \infty$; Merton $J \sim \mathcal{N}(m, \delta^2)$ the running instance.

What is general and what is not

Theorems 1, 2, 4 hold for arbitrary ν ; Proposition 1 and Theorem 3 need symmetry; Gaussianity is not assumed anywhere.

Theorem 1 — the zero-free decomposition

In the continuous-arbitrage limit ($\gamma = 0$, $\Delta t \rightarrow 0$), $P_t \equiv S_t$ and LVR is the Itô–Lévy remainder of $V(S_t)$:

$$\ell = \underbrace{\frac{\sigma^2}{8} V(S_t)}_{\text{diffusion-LVR}} + \underbrace{\frac{\lambda}{2} V(S_t) \mathbb{E}[(e^{J/2} - 1)^2]}_{\text{jump-LVR}}$$

For Merton, $\mathbb{E}[(e^{J/2} - 1)^2] = e^{m+\delta^2/2} - 2e^{m/2+\delta^2/8} + 1 \geq 0$, zero iff $J \equiv 0$.

What to watch

This is the $\Delta t \rightarrow 0$ limit, so *neither* term shows a Δt yet. What matters is where each comes from: the first from the variance of a continuous path, the second from the jump law alone. Put blocks back in and only the first picks up a discount.

Measure

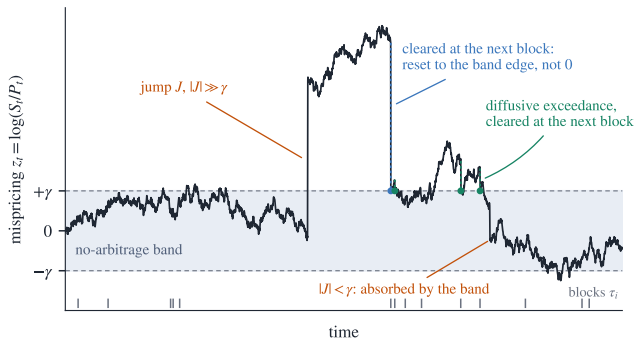
The drift cancels between the two terms, so no normalisation of μ is needed — but with jumps the rate is *not* measure-independent. This is a physical-measure statement, calibrated against realised prices.

The mispricing process

A fee opens a gap between
reference and pool:

$$Z_t := \log(S_t/P_t).$$

- **No-arbitrage band** $|z| \leq \gamma$:
inside it, no trade covers the fee.
- At a block, trade iff $|z| > \gamma$ —
and only out to the *near band edge*, not to 0.



The two channels, separately

Diffusion (Lemma 1). At $\lambda = 0$ the stationary law is uniform on the band with exponential tails, so

$$\ell_{\text{diff}} = \frac{\sigma^2}{8} V \cdot F(\kappa), \quad F(\kappa) = \frac{1}{1 + \sqrt{2} \kappa}$$

with $\kappa = \gamma/(\sigma\sqrt{\Delta t})$; $F(\kappa) = \mathbb{P}_\pi(|z| > \gamma)$, strictly decreasing, $F(\kappa) \rightarrow 0$.

Jumps (Lemma 2). Each jump is charged once, on its excess over the band, at rate λ :

$$\ell_{\text{jump}} = \lambda V \cdot G_\nu(\gamma),$$
$$G_\nu(\gamma) = \frac{1}{8} \mathbb{E}_\nu[(|J| - \gamma)^2 \mathbf{1}\{|J| > \gamma\}]$$

a functional of ν alone. For symmetric Merton, $G = \frac{\delta^2}{8} \Psi(\gamma/\delta)$.

The fee discount is different on each channel

Both channels are trimmed by the fee — but by $F(\gamma/\sigma\sqrt{\Delta t})$ on one and by $\Psi(\gamma/\delta)$ on the other. Only the first contains Δt , because the size of a jump is set by δ , not by how long the block lasted.

Theorem 2 — the separated rate

I define the separated rate ℓ_0 and write the LVR rate as the sum of the separated rate and an error term $\mathcal{E}(\Delta t)$.

$$\ell(\Delta t) = \ell_0(\Delta t) + \mathcal{E}(\Delta t), \quad \ell_0(\Delta t) := \frac{\sigma^2}{8} V \cdot F\left(\frac{\gamma}{\sigma\sqrt{\Delta t}}\right) + \lambda V \cdot G_\nu(\gamma).$$

The error $\mathcal{E}(\Delta t)$ captures a number of effects that are not in the separated rate. This term needs to be bounded to make the decomposition useful, which is what Proposition 1 does.

Proposition 1 Under a mixing condition, the error term $\mathcal{E}(\Delta t)$ is bounded by closed-form expressions from below and above;

- The mixing condition is numerically verified for symmetric jump laws and relevant regimes.
- The bounds derive from six terms: interaction, block clock, single-jump thinning, multi-jump, curvature residual, stationary law.
- The bracket is tight — under 0.29% of the rate across the empirically relevant regime.

Theorem 3 — an exact floor

Let ν be symmetric and satisfy $\mathbb{E}[g(\sum_1^k J)] \geq k \mathbb{E}[g(J)]$ for every k (Merton with $m = 0$ does). Then for every $\Delta t > 0$

$$\ell(\Delta t) \geq \lambda V G_\nu(\gamma) = \inf_{\Delta t' > 0} \ell_0(\Delta t') > 0$$

Corollary 1. ℓ_0 is strictly increasing and the floor is approached only as $\sqrt{\Delta t}$: halving the block time removes at most 29% of the diffusion term.

Key takeaway

The jump channel establishes a strictly positive floor to the LVR rate, beyond which block-time reductions do not help. Approaching the floor is slow, at the rate $\sqrt{\Delta t}$.

The planner's optimum, and the numbers

The planner's welfare problem

Shorter blocks are always better, down to the floor — so why not $\Delta t \rightarrow 0$? Because blocks are not free.

- Each block costs fixed work whatever it carries — gossip, signature aggregation, the commit round — so the chain-level rate is $R(\Delta t) = r/\Delta t$.
- That cost is not observable, so I proxy r by issuance: it overstates where it subsidises growth and understates where fees and MEV pay validators.
- A pool of TVL V inherits a share by value secured: $c/\Delta t$ with $c := r V/V_{\text{chain}}$.

$$\min_{\Delta t > 0} W(\Delta t) = \underbrace{\ell_0(\Delta t)}_{\text{LVR}} + \underbrace{c/\Delta t}_{\text{cost of consensus}}$$

The trade-off

Two terms, two forces: the LVR pulls to the floor, the other grows without bound.

Theorem 4 — netting LVR against what a block costs

Minimising W gives a first-order condition in $\eta = \sqrt{2} \kappa$:

$$\eta^{\text{opt}}(1 + \eta^{\text{opt}})^2 = \frac{V\gamma^2}{8c}, \quad \Delta t^{\text{opt}} = \frac{2\gamma^2}{\sigma^2(\eta^{\text{opt}})^2},$$

a cubic with a unique positive root, closed form by Cardano.

Importantly, the optimum is a property of the block time and the fee tier only;

- $\partial \Delta t^{\text{opt}} / \partial \lambda = \partial \Delta t^{\text{opt}} / \partial m = \partial \Delta t^{\text{opt}} / \partial \delta = 0$: the jump parameters do not enter the first-order condition.
- $\partial \Delta t^{\text{opt}} / \partial V = 0$: the pool size does not enter either.

What the remainder does to it

That invariance is a property of ℓ_0 ; $\mathcal{E}$ carries both λ and Δt . Bounding it directly puts any exact minimiser in $[7.3, 9.9]$ s — the range over which W is within 1 bp/yr of its minimum anyway.

Calibration

Price. Binance ETH/USDT 5-minute closes, Jan 2020 – Jun 2026 (2,373 days, 682,946 bars):

- $\sigma = 0.8156$, the continuous part of realised variance.
- $\lambda = 283.3/\text{yr}$ and $\delta = 0.0192$, from Lee–Mykland jump detection.
- $m = 0$: the sample gives $\hat{m} = -0.0012$, with a 99% interval that contains zero.

Consensus cost. Ethereum's own issuance, as a proxy for the resource cost of producing a block:

- Gross issuance $\sim \$1.7\text{B}/\text{yr}$ against $V_{\text{chain}} \approx \208B , scaled by 0.83 because staked supply ($\sim 30\%$) sits above Drake's security-optimal quarter.
- Attributed to the pool pro-rata by value secured, $V/V_{\text{chain}} = 4.8 \times 10^{-6}$, giving $c \approx \$260 \times 10^{-5}$ per block.

Modelling choices. $\gamma = 5$ bp (Uniswap 0.05% tier), $V = \$1\text{M}$.

The rate across deployed block times — and the optimum

Chain	Δt	ℓ (bp/yr)	diff. share
Ethereum L1	12 s	471	73%
Base / OP L2	2 s	312	60%
Solana	400 ms	221	43%
Arbitrum	250 ms	203	38%
App-chain	50 ms	162	23%
Jump floor	—	125	0%

- Ceiling $\sigma^2 V/8 = 832$ bp/yr; floor $\lambda VG = 125$.
- A $240\times$ cut, $12\text{ s} \rightarrow 50\text{ ms}$, removes only 89% of the diffusion term and leaves the rate 29% above the floor.
- The channels are equal at $\Delta t_x \approx 0.75\text{ s}$.

Where the optimum lands

Theorem 4 gives $\Delta t^{\text{opt}} \approx 8.4\text{ s}$ — two-thirds of Ethereum's slot, and far above sub-second timing. The gain is small and the optimum shallow: W falls $539 \rightarrow 533$ bp/yr, under 2%, and stays within 1 bp/yr of its minimum over $[7.4, 9.7]\text{ s}$.

What this means, and what it does not

Policy

- Price jumps set a floor beyond which block time reductions do not increase LP welfare.
- The diffusion channel stays material into the low-millisecond range — so the block time optimum is bound by the cost of a block, not by the jump channel.
- The block time optimum is invariant in pool size and jump regime.
- Once the floor does bind, the remaining LP-side levers are not the schedule:
 - ▶ fee-tier design — higher γ reduces both channels, but also reduces trade volume, price accuracy and fee revenue (both LP and protocol);
 - ▶ batch auctions — clear a batch at one uniform price; no stale quote left standing to race against;
 - ▶ auction-managed AMMs — keep the quote, but sell the right to arb it; the rent goes to LPs.

Back to the two proposals

Arbitrum, 2 s $\rightarrow$ 250 ms: the model gives 312 $\rightarrow$ 203 bp/yr, a 35% cut (not 65%).

EIP-7782, 12 s $\rightarrow$ 6 s: the model gives 471 $\rightarrow$ 403 bp/yr, a 14% cut (not 29%).

Both fail to price the jump channel.

Conclusion

What the model says

- 125 bp/yr — the jump floor. Exact for symmetric ν , at every block time.
- 471 bp/yr at Ethereum's 12 s, 73% of it in the channel the schedule governs — which yields only as $\sqrt{\Delta t}$.
- 8.4 s — the LP-side optimum, invariant in pool size and in the jump parameters.

What it's limitations are

- The objective is LP-LVR only: Other factors like finality, UX, censorship resistance, MEV are not captured.
- Issuance is a proxy for block production cost.
- The arbitrageur is idealised; blocks are Poisson; the pool is full-range.

Key takeaway

The LVR rate splits into a channel the schedule governs and one it cannot touch. The jump floor is genuine but distant: what binds the block time optimum is the cost of a block, and the lever against it is γ .

Questions and discussion.

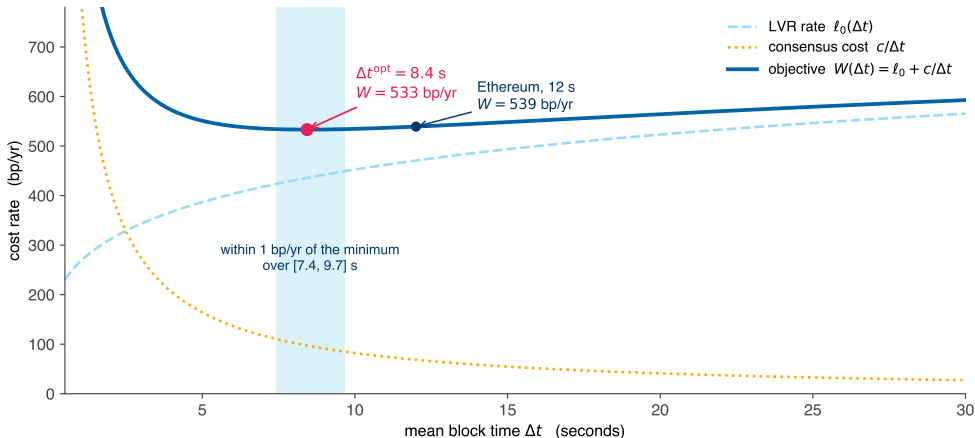
Zurich University of
Applied Sciences

ZHAW School of Engineering

Nils Bundi
Senior Lecturer
Technikumstrasse 9 | PO Box
8401 Winterthur, Switzerland
bund@zhaw.ch
www.zhaw.ch/engineering

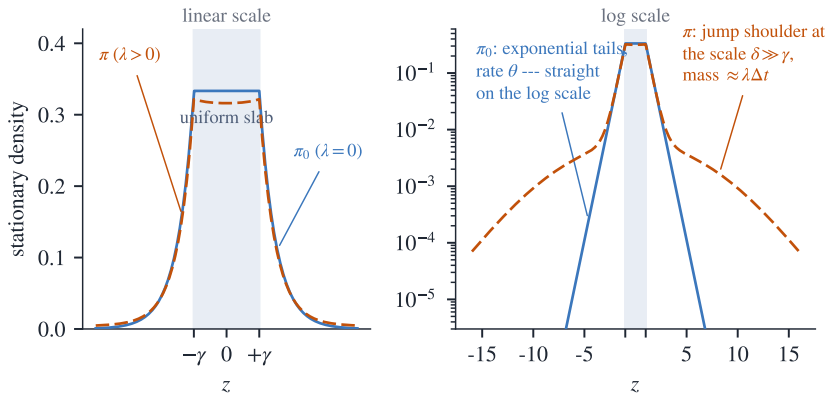
Backup

The objective, and its shallow optimum



The optimum is shallow: 539 $\rightarrow$ 533 bp/yr. Across plausible c it runs 0.2–15 s; the jump parameters move it not at all.

The stationary law of the mispricing



Pre-block mispricing at $\eta = 2$, $\delta = 5\gamma$, $\lambda\Delta t = 0.08$. Without jumps π_0 is uniform on the band with exponential tails; jumps add a far field at the jump scale $\delta \gg \gamma$ — the shoulder leaving the tail on the log scale, mass no block schedule can pull in.